

1. Traffic Surveillance Model Selection

Given Requirements

- Accuracy $\geq 90\%$
- Latency ≤ 80 ms
- Works under varying lighting

Method	Accuracy	Latency	Result
A	88%	60 ms	✗ Accuracy too low
B	91%	85 ms	✗ Latency too high
C	92%	75 ms	✓ Satisfies both

Answer:

Method C is the best choice. It provides **92% accuracy** and **75 ms latency**, satisfying both required constraints. Method A fails the accuracy requirement, while Method B fails the latency requirement.

Decision: Method C.

2. Drone Vision System Constraint

Requirements

- Lightweight computation
- Real-time detection < 60 ms
- Accuracy $\geq 85\%$

Model	Accuracy	Latency	Result
A	83%	40 ms	✗ Accuracy too low
B	87%	65 ms	✗ Too slow
C	86%	55 ms	✓ Satisfies requirements

Answer:

Model C is the most suitable choice. It provides **86% accuracy** and **55 ms detection time**, satisfying both accuracy and real-time requirements. It is also more suitable for a drone because it has a reasonable computational requirement.

Decision: Model C.

3. Security System Recognition

Requirements

- Accuracy > 92%
- Moderate computation

Approach	Accuracy	Computation	Result
Traditional	85%	Low	✗ Accuracy too low
Deep Learning	94%	High	⚠ High computation
Hybrid	91%	Moderate	✗ Accuracy too low

Answer:

Deep Learning is the best choice because it provides **94% accuracy**, which satisfies the accuracy requirement. Although its computation is high, it provides the required recognition performance.

The Hybrid approach has moderate computation but only **91% accuracy**, which does not meet the required >92%.

Decision: Deep Learning, especially when security and accuracy are more important than computational cost.

4. Motion Tracking in Sports Analytics

Requirements

- Accurate motion tracking
- Real-time performance

Comparison

Method	Accuracy	Speed
Optical Flow	High	Slow
Motion Estimation	Lower	Fast

Answer:

Motion Estimation is the more suitable choice for real-time sports analytics because it is fast and can provide immediate tracking results.

Although Optical Flow provides better accuracy, its slow processing may not satisfy real-time requirements.

Decision: Motion Estimation when real-time performance is the main constraint. If maximum tracking accuracy is more important and some processing delay is acceptable, Optical Flow can be preferred.

5. Face Recognition System Design

Requirements

- Accuracy $\geq 90\%$
- Works in low light
- Moderate computation

Method	Accuracy	Result
Traditional	82%	✗ Does not meet accuracy
Deep Learning	93%	✓ Meets accuracy
Hybrid	89%	✗ Below required accuracy

Answer:

Deep Learning is the best choice because it provides **93% accuracy**, satisfying the $\geq 90\%$ requirement. Deep-learning-based face recognition can also be trained with low-light and varied-lighting images, improving robustness under difficult lighting conditions.

Although its computational requirement may be higher than traditional methods, it provides the required accuracy and robustness.

Decision: Deep Learning.